

&" NuclearCraft

Plutonium Age

Complete Player Guide

Items · Recipes · Machines · Radiation · Zombies · Boss

Material Substitution Reference

Custom items cannot be placed directly in crafting grids. Use these vanilla materials as proxies — the plugin validates the real custom item automatically.

Vanilla Material (place in grid)	Represents (custom item)	CMD
Echo Shard	Refined Plutonium Ingot	1103
Yellow Wool	Industrial Fabric	1315
Magma Cream	Radioactive Core	1101
Prismarine Crystals	Raw Plutonium Fragment	1102
Heart of the Sea	Irradiated Heart	1106
Pink Petals	Healing Petal	1105
Wheat Seeds	Mutated Seed	1104
Blast Furnace (block)	Nuclear Smelter (machine)	1201
Smithing Table (block)	Nuclear Forge (machine)	1401
Diamond Pickaxe	Radiation Drill	1108
Leather armor pieces	Hazmat Suit pieces	1306–1309
Netherite armor pieces	Plutonium Armor pieces	3001–3004
Honey Bottle	Radiation Antidote	1301
Glass Bottle	Radiation Serum	1302

All Custom Items

Materials & Components

Item	Base Material	CMD	Notes
Raw Plutonium Fragment	Prismarine Crystals	1102	Mined from ore. Smelter input.
Refined Plutonium Ingot	Echo Shard	1103	Smelter output. Main crafting mat. Repairs plut. reactor (1,000 dur).
Radioactive Core	Magma Cream	1101	Zombie drop. Forge fuel (2,000 energy). High radiation.
Irradiated Heart	Heart of the Sea	1106	Rare zombie drop. MK-3/4 upgrade + Titan armor.
Industrial Fabric	Yellow Wool	1315	Crafts & repairs Hazmat Suit (+200 dur per anvil use).
Mutated Seed	Wheat Seeds	1104	Zombie drop. Plant on Radioactive Farmland only.
Healing Petal	Pink Petals	1105	Farm harvest. Antidote/serum ingredient.
Titan Core	Nether Star	1107	Direct: -25 rad + Regen II. Dropped by Titan or crafted (Phase 11). 4 needed to summon Titan.
Titan Fragment	Amethyst Shard	1108	Titan loot. High-tier crafting (Phase 11+).
Reactor Heart	Nether Star	1109	Rare Titan drop (10%). MK-3/4 upgrade ingredient.
Mutated Crystal	Amethyst Shard	1111	Titan drop (8%). Future crafting.
Ancient Reactor Blueprint	Paper	1110	Very rare Titan drop (5%). Phase 11+ unlock.

Machines, Tools & Consumables

Item	Base Material	CMD	Function
Nuclear Smelter	Blast Furnace	1201	Refines Raw Fragments !' Ingots. Emits 8 rad/5s within 3 blocks.
Nuclear Forge	Smithing Table	1401	Upgrades gear MK-1! MK-4. Radioactive Core fuel.
Radiation Drill	Diamond Pickaxe	1108	Only safe Plutonium Ore mining tool. ×1.2 speed. 1,800 dur.
Radiation Antidote	Honey Bottle	1301	Right-click: clears ALL radiation & debuffs. No immunity.
Radiation Serum	Glass Bottle	1302	Right-click: clears ALL radiation + 10 min immunity.

Crafting Recipes

P = Echo Shard (Refined Plutonium Ingot) W = Yellow Wool (Industrial Fabric) S = Stick R = Magma Cream (Radioactive Core)

Machine Recipes

Nuclear Smelter

Diamond	Obsidian	Diamond
Obsidian	Furnace	Obsidian
Diamond	Obsidian	Diamond

! D=Diamond O=Obsidian F=Furnace

Nuclear Forge

Obsidian	Anc.Debris	Obsidian
P (Ingot)	Sm.Table	P (Ingot)
Obsidian	Anc.Debris	Obsidian

! P=Echo Shard (Ref.Ingot proxy)

Plutonium Block

P	P	P
P	P	P
P	P	P

! 9× Refined Plutonium Ingots ! compact storage

Lead-Lined Crate

Iron	Iron	Iron
Iron	Chest	Iron
Iron	Iron	Iron

! Stores fragments with zero radiation leak

Radiation Drill

Radiation Drill — only tool that safely mines Plutonium Ore

R-Core	Diamond	R-Core
Diamond	Diamond	Diamond
	Stick	

! R-Core=Radioactive Core (Magma Cream proxy) · ×1.2 speed · 1,800 dur

Plutonium Weapons & Tools

Plutonium Sword (+10 DMG, +10 rad/hit, crit +5 rad, 2,500 dur)

	P	
	P	
	S	

Plutonium Axe (+11 DMG, 15% shockwave 3-block +15 rad, 2,600 dur)

P	P	
P	S	
	S	

Plutonium Pickaxe (10% debris find chance, 2,800 dur)

P	P	P
	S	
	S	

Plutonium Shovel (15% soil!'Radioactive Farmland, 2,500 dur)

	P	
	S	
	S	

Plutonium Hoe (2,000 dur)

P	P	
	S	
	S	

Plutonium Arrow (×4 output — +25 rad, Poison I 4s, Glowing 5s)

	P	
	Arrow	
	Feather	

!' P=Echo Shard · Full charge crit: +20 bonus radiation

Armor & Cure Recipes

Plutonium Armor (P = Echo Shard proxy)

All pieces: Protection III + Unbreaking III · Full 4-piece set: environmental radiation IMMUNITY + Speed II + Fire Resistance

Plutonium Helmet (4A 2T 450dur -15%rad)

P	P	P
P		P

Plutonium Chestplate (8A 2T 640dur -15%rad)

P		P
P	P	P
P	P	P

Plutonium Leggings (6A 2T 590dur -15%rad)

P	P	P
P		P
P		P

Plutonium Boots (4A 2T 480dur -15%rad Feather Fall II)

P		P
P		P

Hazmat Suit (W = Yellow Wool proxy)

Full 4-piece set: 80% radiation reduction · Leather base · Repair with Industrial Fabric

Hazmat Helmet (2A 0.5T 363dur -20%rad)

W	W	W
W		W

Hazmat Chestplate (5A 0.5T 529dur -30%rad)

W		W
W	W	W
W	W	W

Hazmat Leggings (4A 0.5T 496dur -20%rad)

W	W	W
W		W
W		W

Hazmat Boots (2A 0.5T 430dur -10%rad)

W		W
W		W

Cure Recipes

Radiation Antidote (SHAPELESS — any slot order)

Healing Petal	Healing Petal	Honey Bottle
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! Clears ALL radiation, debuffs & infection. No immunity granted.

Radiation Serum

Heal.Petal	R-Core	Heal.Petal
Gold Nugget	Gold Apple	Gold Nugget
Heal.Petal	Glass Bottle	Heal.Petal

! R-Core=Radioactive Core · Clears ALL radiation + 10 min full immunity

Nuclear Smelter Processing

Input	! Raw Plutonium Fragment	Output	1× Refined Plutonium Ingot
Processing time	15 seconds (300 ticks)		
Fuel required	Coal (100u) · Charcoal (80u) · Coal Block (900u) · Blaze Rod (120u) · Lava Bucket (1,000u) • best)		
Min temperature	500°C (machine heats up first, then processes)		

Radiation System

Stages

Stage	Name	Points	Effects
0	Healthy	0–99	None
1	Minor Exposure	100–249	Weakness I · Nausea
2	Moderate Exposure	250–499	Weakness I · Slowness I · Nausea · 5% proximity spread
3	Severe Exposure	500–749	Weakness II · Slowness II · Hunger I · -0.5 HP/cycle · 15% spread
4	Critical Poisoning	750–1000	Weakness III · Slowness III · Hunger II · -1 HP/cycle · 25% spread

Key Radiation Sources

Source	Amount	Trigger
Plutonium Ore "d1 block	+5 pts	every 5s
Plutonium Ore "d2 blocks	+2 pts	every 5s
Mining ore WITHOUT Radiation Drill	+25 burst	per attempt (5s cooldown)
Radioactive Debris (5-block radius)	+0.5/s	continuous
Nuclear Smelter active (3-block radius)	+8 pts	every 5s
Radiation Cloud (20% on zombie death)	+5/s	10s, 3-block radius
Irradiated Zombie Lv.1–4 hit	+10/+20/+35/+50	per melee hit by level
Titan — Radiation Aura	+3/s	10-block radius, all phases
Titan — Radiation Wave	+40 pts	20-block radius burst (Phase 2+)
Titan — Reactor Overload	+70 pts	18-block radius (Phase 3+)
Titan — Nuclear Catastrophe	+80 pts + sets HP	30-block radius (Phase 4)
MK-IV Forge Overload	to 1 +150 pts	8-block radius burst

Protection & Cures

Protection Method	Radiation Reduction
Generic armor — per piece	-15% per piece (max -60% full set)
Full Hazmat Suit (4/4)	-80% total
Full Plutonium Armor (4/4)	IMMUNE — zero environmental radiation
Plutonium Boots special	No ground-contact radiation
Near ore with Hazmat suit	×0.20 multiplier (80% blocked)
Near ore with Plutonium armor	×0.00 (complete immunity to ore zone)

Contagion (Spread Between Players)

Method	Condition	Amount	Chance
Proximity ("d3 blocks)	Infected Stage 2	+25 pts	5% / 5s
Proximity ("d3 blocks)	Infected Stage 3	+25 pts	15% / 5s
Proximity ("d3 blocks)	Infected Stage 4	+25 pts	25% / 5s
Melee hit (physical contact)	Any stage	+50 pts	35% / hit
Shared vehicle (boat/cart)	Any stage	+25 pts	20% / 15s

Natural decay	Requires 10 min clean !' -2 pts per 60s
Radiation Antidote	Clears ALL radiation + debuffs instantly (no immunity)
Radiation Serum	Clears ALL + grants 10 min full immunity

Machines — Nuclear Smelter & Nuclear Forge

Nuclear Smelter — Usage

1. Craft Smelter (Diamond+Obsidian+Furnace pattern)
2. Place block, right-click to open GUI
3. Add Raw Plutonium Fragments (Input) and fuel (Fuel slot)
4. Machine heats to 500°C then processes — collect Refined Ingots from Output

Fuel	Units	Duration
Coal	100u	~20s
Charcoal	80u	~16s
Coal Block	900u	~3 min
Blaze Rod	120u	~24s
Lava Bucket	1,000u	~3m20s !* most efficient per slot

&" Active smelter emits +8 radiation every 5 seconds within 3 blocks. Keep it in a sealed room.

Nuclear Forge — Upgrade Tiers

Place block, insert Radioactive Cores (fuel), place item + materials, click UPGRADE.

Tier	Ingots	Cores	Hearts	Energy	Success	Bonus
MK-1	2	1	0	500	100%	+5% dmg/speed/armor/dur
MK-2	4	2	0	1,200	90%	+10% all stats
MK-3	8	4	1	2,500	75%	+20% all stats
MK-4	16	8	2	5,000	50%	+35% all stats · 5% downgrade on fail

&" MK-IV equips a Radiation Aura: +25 rad to nearby mobs, +15 to enemy players, every 2s within 3 blocks.

&" Forge Overload (>10,000 energy): +150 radiation burst in 8-block radius — never load more than 5 Cores.

Ore, Mining & Radioactive Farming

Plutonium Ore

Parameter	Value
Y-level	" 64 to " 58 (deepest deepslate layer, Overworld only)
Spawn rate	60% of newly generated chunks · Vein size: 1–2 blocks
Mine WITHOUT Radiation Drill	+25 radiation burst · Ore gives nothing · 5s cooldown
Mine WITH Radiation Drill	1 Raw Plutonium Fragment (Fortune I: 1–2, II: 1–3, III: 1–4) · 3–7 XP
Near ore passive radiation	"d1 block: +5/5s · "d2 blocks: +2/5s · "d3 blocks: +1/5s
Hazmat suit near ore	80% radiation reduction (×0.20 multiplier)
Plutonium armor near ore	Complete immunity (×0.00 multiplier)

Radioactive Farming

Steps: Get Mutated Seeds from zombies !' dig soil with Plutonium Shovel (15% farmland conversion) !' plant seeds on Radioactive Farmland !' harvest at Stage 4 (5 stages total) !' get 1–3 Healing Petals + 0–2 bonus seeds.

Parameter	Value
Radioactive Farmland growth bonus	+50% extra growth chance per random tick
Harvest drops	1–3 Healing Petals · 0–2 Mutated Seeds · 2–5 XP
Toxic Bloom chance	1% — fully grown crop becomes a radiation hazard instead of dropping
Toxic Bloom danger	+8 radiation every 2s in 4-block radius — break immediately
Farmland passive radiation	+1 pt / 2s to nearby players (Plutonium Boots negate this)

Irradiated Zombies

60% of natural zombie spawns become Irradiated. Baby zombies excluded.

Level Stats & Loot

Level (spawn%)	HP	DMG	Speed	KB Res	Rad/Hit	XP	Core%	Heart%
Lv.1 Standard (80%)	35	5	×1.15	0.15	+10	10	15%	1%
Lv.2 Enhanced (15%)	45	7	×1.20	0.20	+20	20	25%	3%
Lv.3 Powerful (4%)	60	9	×1.25	0.30	+35	40	40%	7%
Lv.4 Alpha (1%)	80	12	×1.30	0.50	+50	100	100%	25%
&" Level 4 Alpha: green Bossbar, Glowing effect. Also drops 50% Mutated Seeds. Treat as a mini-boss.								
&" Death cloud: 20% chance per zombie !' +5 rad/s, 3-block radius, 10 seconds. Move away immediately.								
&" Radiation Surge: 5% chance per night — doubles all radiation AND all loot for the entire night.								

Plutonium Titan Boss

Summoning Ritual

ALTAR: Crying Obsidian at center. Obsidian blocks at 4 corners (2 blocks diagonal from center, same Y).
REQUIRED: 4× Titan Cores (consumed from inventory) + 1× Irradiated Heart (in hand, right-click center block).
ANIMATION: 15 seconds !' Titan spawns. COOLDOWN: 60 minutes. Only one Titan at a time.

Health & Phases

Players Nearby		Titan HP	
1–3		5,000 HP	
4–6		6,500 HP	
7–10		8,500 HP	
10+		10,000 HP	
Phase (HP%)	Spd	DMG	Abilities Active
Phase 1 (100–75%)	×1.00	×1.00	Radiation Aura · Titan Slam
Phase 2 (75–50%)	×1.10	×1.20	+ Radiation Wave · Mutant Summon (3+1Alpha)
Phase 3 (50–25%)	×1.25	×1.50	+ Reactor Overload · Energy Beam
Phase 4 (25–0%)	×1.50	×2.00	+ Nuclear Catastrophe · Final Frenzy (+50% ATK speed, one-time)
&" Nuclear Catastrophe (Phase 4): 10s charge !' 30-block radius !' +80 radiation + sets all players to 1 HP. Retreat!			

Titan Rewards

Reward	Chance / Amount
Titan Fragments	Guaranteed · 1–8+ by contribution
Refined Plutonium Ingots	Scales with contribution %
Reactor Heart	10% per qualifying player
Ancient Reactor Blueprint	5% per qualifying player
Mutated Crystal	8% per qualifying player
Loot Chest at death location	Always — Cores, Fragments, Ingots inside
Server XP	5,000 XP total, split by contribution

Combat System & Admin Commands

Weapon Effects & Mastery

Weapon / Hit	Radiation	Extra Effect
Plutonium Sword base hit	+10 rad	—
Sword crit (falling attack)	+15 rad	+5 bonus on crit
Sword vs Stage 3+ victim	+15 rad	+5 bonus vs infected
Plutonium Axe shockwave (15%)	+15 rad AoE	3-block radius burst
Plutonium Arrow hit	+25 rad	Poison I (4s) · Glowing (5s)
Arrow — full charge crit	+45 rad	+20 bonus on max draw
Combo system	Up to 8-hit stack · +5 rad per stack · max +35 bonus · resets after 6s idle	

Weapon Mastery	Novice!'Experienced(100XP)!'Veteran(500)!'Elite(1,500)!'Master(4,000): at Master +10% all weapon radiation
Repair Plutonium gear	Anvil + Echo Shard (Refined Ingot) !' +300 durability per ingot
Repair Hazmat Suit	Anvil + Yellow Wool (Industrial Fabric) !' +200 durability per piece

Admin Commands (/nuclearcraft or /nc)

Command	Description
/nc radiation set <player> <0-1000>	Set a player's radiation level
/nc radiation add <player> <amount>	Add radiation to a player
/nc radiation clear <player>	Clear all radiation
/nc radiation status [player]	Check radiation stage and points
/nc ore give fragment [amt]	Give Raw Plutonium Fragments
/nc ore give drill	Give a Radiation Drill
/nc equipment give <type>	Give plutonium/hazmat gear piece
/nc farming give <seed petal antidote serum>	Give farming/cure items
/nc smelter give	Give a Nuclear Smelter block
/nc forge give	Give a Nuclear Forge block
/nc titan spawn	Force-spawn the Titan at your location
/nc titan kill	Despawn the active Titan
/nc titan phase <1-4>	Force Titan to a specific phase
/nc reload	Reload all YAML configs without restart

